# Eligibility Form: IS ASYMPTOMATIC ATHEROSCLEROSIS IN CAROTID ARTERIES ASSOCIATED WITH COGNITIVE FUNCTION? A SYSTEMATIC REVIEW AND META-ANALYSIS

| **1. Reference details** | | | |
|---|---|---|---|
| Reference citation | | | |
| First author | | | |
| Year of publication | | | |
| Title of the paper | | | |
| Assessor's identifier | 1 | 2 | |
| **2. Study eligibility** | | | |
| Inclusion of the study | Yes √ | No √ | Unclear r |
| **Eligibility criteria** | **Reason for exclusion (if excluded)** | Yes | No |
| ALL studies except : review articles, conference abstract, letters to the editor, case reports; pilot study, | ☐ no MMSE cognitive test | ☐ | ☐ |
| cIMT measured by ultrasound | ☐ No IMT Assessment | √ | ☐ |
| | ☐ Other imaging modality not ultrasound | ☐ | ☐ |
| | ☐ review articles, conference abstract, letters to the editor, case reports | ☐ | ☐ |
| Notes: | ☐ Duplicate | ☐ | ☐ |
| | | | |
| **Additional information** | | | |
| Notes: | | | |

DO NOT PROCEED IF PAPER EXCLUDED FROM REVIEW

## Data Extraction Form

| **3. Study details** | | |
|---|---|---|
| Study (cohort) name | The EVA Study | |
| Study design | longitudinal study | |
| Region/country | Nantes (western France). | |
| Sample size | 1279 | |
| Mean (range) age | Mean ± SD :<br>mean age, 65.0+/-3.0 years | Range (IQR): |
| Sex | Male, n (%):<br>526 men | Female, n (%): 753 women |
| Follow-up duration | | |
| Clinical variables (n, %) | | |
| ☐ Hypertension | n./a | |
| ☐ Diabetes mellitus | n/a | |
| **4. Exposure details** | | |
| Hardware: Ultrasound Machine | an Aloka SSD-650 with a 7.5-MHz transducer. | |
| Method and site | The ultrasound examination involved scanning the common carotid arteries, the carotid bifurcations, and the first 2 cm of the internal carotid arteries. The IMT (distance between the media-adventitia interface and the lumen-intima interface) was automatically measured twice on longitudinal B-mode images of the far wall of each common carotid artery at plaque-free sites. The common carotid IMT was defined as the mean of four measurements. All segments of the carotid arteries were scanned longitudinally and transversely to assess the presence of plaques, and a localized echo-structure protruding into the lumen was considered to be a plaque if the distance between the media-adventitia interface and the internal side of the lesion was >or=to1 mm. The total number of plaques was recorded. For each, maximum thickness was measured perpendicular to the vessel wall. When several plaques were present in the same carotid segment (ie, common carotid artery or bifureation/origin of the internal carotid artery), plaque thickness measurement was made only for that with the greatest protrusion into the lumen. On the basis of transverse views, the degree of carotid stenosis was defined as (1-residual area/vessel area) times 100%. | |

| Notes: |
|---|

| **5. Outcomes' details** | |
|---|---|
| Type has of outcomes used in this review | Outcomes reported in this study (n) |
| **Primary outcome(s)**<br>Global cognitive function | neuropsychological battery that included seven tests assessing different cognitive functions and a global test (MMSE) of 18 items roughly assessing various cognitive skills with scores ranging from 0 to 30. |
| Notes: | |

| **6. Available number of participants** | | | | | |
|---|---|---|---|---|---|
| | | | | | Number |
| Baseline sample size | Yes | | No | | |
| follow-up | Yes | | No | | |
| Total number included in the analysis | Yes | | No | | |
| All subjects accounted for | Yes | | No | | |
| Notes: | | | | | |

| **7. Statistical analysis** | |
|---|---|
| Statistical method used | Nominal polychotomous logistic regressions [13] (BMDP statistical software) were used to analyze the relationships between each cognitive score distribution of the eight neuropsychological tests and carotid characteristics (presence of plaques, IMT), adjusting for age, education, depressive symptomatology, systolic blood pressure, body mass index, smoking, and alcohol consumption. |